

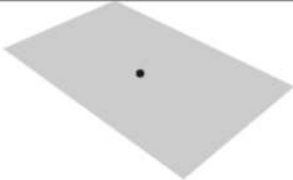
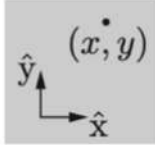
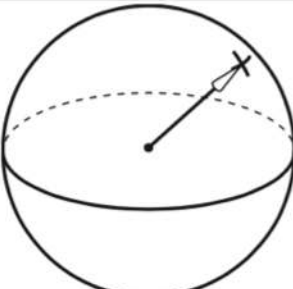
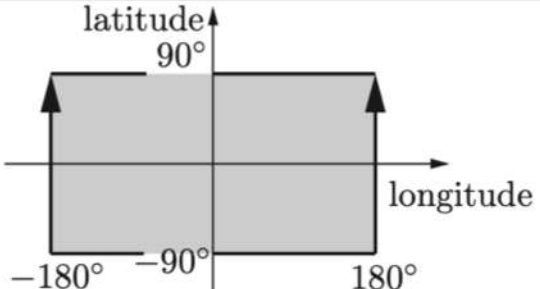
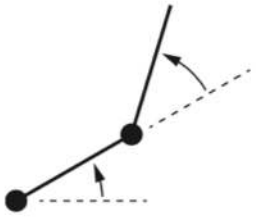
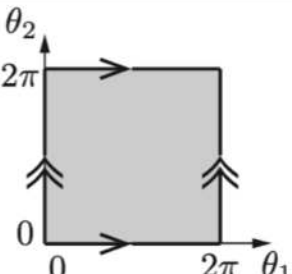
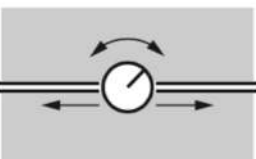
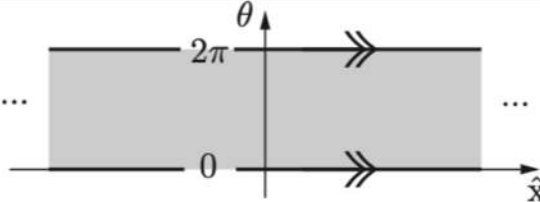
Chapter 2	Configuration Space
	2.1 DOF of a Rigid Body
	2.2 DOF of a Robot
	2.3 C-space Topology and Representation
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

- Two C-spaces may have the same dof but differ in other ways. The **topology** (“shape”) of a space is independent of how we **represent** it.
- Two spaces are **topologically equivalent** if one can be continuously deformed to the other without cutting or pasting.
- Some spaces are **Cartesian products** of spaces of lower dimension, e.g.,

$$\textbf{(1d)} \ \mathbb{E}, S = T \quad \textbf{(2d)} \ \mathbb{E} \times \mathbb{E} = \mathbb{E}^2, S \times S = T^2, S^2, \mathbb{E} \times S \quad \textbf{(higher)} \ \mathbb{E}^k \times S^m \times T^n$$

- Represent Euclidean (“flat”) spaces $\mathbb{E}^n$ as $\mathbb{R}^n$. For curved spaces, choose
 - minimum-parameter **explicit parameterizations** (choose between **singularities** or an **atlas** of **coordinate charts**), OR
 - **implicit representation** (use more numbers subject to constraints).

system	topology	sample representation
 point on a plane	 $\mathbb{E}^2$	 $\mathbb{R}^2$
 spherical pendulum	 S^2	 latitude 90° -90° -180° -90° 180° longitude [−180°, 180°) × [−90°, 90°]
 2R robot arm	 $T^2 = S^1 \times S^1$	 θ_2 2π 0 0 2π θ_1 [0, 2π) × [0, 2π)
 rotating sliding knob	 $\mathbb{E}^1 \times S^1$	 θ 2π 0 ... $\hat{x}$... $\mathbb{R}^1 \times [0, 2\pi)$

Any value in an atlas of coordinate charts?
An implicit representation?

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hexrotor with two 5-DOF arms

<https://www.prodrone.com/archives/1420/>

each arm joint $\in (0, \pi)$

$$\mathbb{R}^3 \times S^2 \times S^1 \times T^6 \times \mathbb{R}^{10} = \mathbb{R}^3 \times S^2 \times T^7$$

C-space topology, with and without
arm joint limits, rotor angles?
Implicit/explicit representations?
Grübler's formula?

$$\text{body: } \mathbb{R}^3 \times S^2 \times S^1$$

$$\text{rotors: } S^1 \times S^1 \dots = T^6$$

6 times

$$\text{arm: } T^5 \text{ (no joint limits)}$$

$$\mathbb{R}^3 \times S^2 \times S^1 \times T^6 \times T^5 \times T^5 =$$

$$\mathbb{R}^3 \times T^{17} \times S^2$$



KUKA youBot
 mecanum-wheel omnidirectional base
 moving on flat ground
 plus 5-DOF robot arm + gripper

C-space topology and representation?
 Include gripper, wheel angles?

$$\underbrace{\mathbb{R}^2 \times S^1}_{\text{chassis}} \times \underbrace{\mathbb{R}^5}_{\text{arm}} \times \underbrace{T^4}_{\text{wheels}}$$

or, if joint limits

$$[a, b]^5$$

$$I^5$$